

Automatic Speech Recognition Benchmarking in Indian Languages

A Comparative Study of Open-Source and Commercial ASR Systems

Hindi · Bengali · Marathi · Telugu · Tamil

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1. Introduction

1.1 Motivation

India is among the most linguistically diverse nations on Earth, with over 1,600 spoken languages and 22 constitutionally recognised ones. Despite this richness, Automatic Speech Recognition (ASR) technology has historically been dominated by English-centric systems. The result is a deep digital divide: hundreds of millions of speakers of languages such as Hindi, Bengali, Marathi, Telugu, and Tamil lack access to reliable voice-driven technology in their native tongue.

The rapid proliferation of smartphones and affordable internet in India makes ASR a critical enabler of digital inclusion. Voice search, accessibility tools, transcription services, and conversational AI all depend on accurate ASR. Benchmarking the current state of ASR in major Indian languages is therefore both timely and practically important.

1.2 Objectives

This report aims to:

- Survey the top five most widely spoken Indian languages and the ASR models available for them.
- Review speech signal processing fundamentals relevant to ASR.
- Trace the evolution of ASR technology from classical probabilistic models to modern foundation models.
- Identify and compare the top state-of-the-art (SOTA) ASR models for each language.
- Evaluate performance using standardised metrics (WER, CER) on publicly available benchmarks.
- Discuss findings and outline future research directions.

1.3 Scope and Limitations

This report focuses exclusively on the five most spoken Indian languages by native-speaker count: Hindi, Bengali, Marathi, Telugu, and Tamil. Results are based on publicly reported benchmarks and reproducible inference on standard hardware. Real-time streaming ASR, dialect variation, code-switching, and domain-specific speech (medical, legal) fall outside the scope of this study.

2. Top Five Spoken Languages in India

The five languages studied in this report collectively account for more than one billion speakers. Table 1 summarises their key characteristics.

Table 1. Key characteristics of the five languages studied

Language	Native Speakers	Language Family	Script	Primary Region
Hindi	553 million	Indo-Aryan	Devanagari	North and Central India
Bengali	234 million	Indo-Aryan	Bengali	West Bengal, Bangladesh
Marathi	99 million	Indo-Aryan	Devanagari	Maharashtra
Telugu	96 million	Dravidian	Telugu	Andhra Pradesh, Telangana
Tamil	81 million	Dravidian	Tamil	Tamil Nadu, Sri Lanka

3. Speech Signal Processing Fundamentals

3.1 The Raw Speech Signal

A raw speech signal is the original audio waveform obtained directly from a speaker's voice through a recording device, prior to any preprocessing or feature extraction. It is represented as a sequence of amplitude values that vary with time and contains all speech-related information, such as linguistic content, pitch, tone, pauses, and background noise. In speech processing and ASR systems, the raw speech signal serves as the initial input for further processing and feature extraction techniques, such as Mel-frequency cepstral coefficients (MFCCs) and Mel spectrograms.

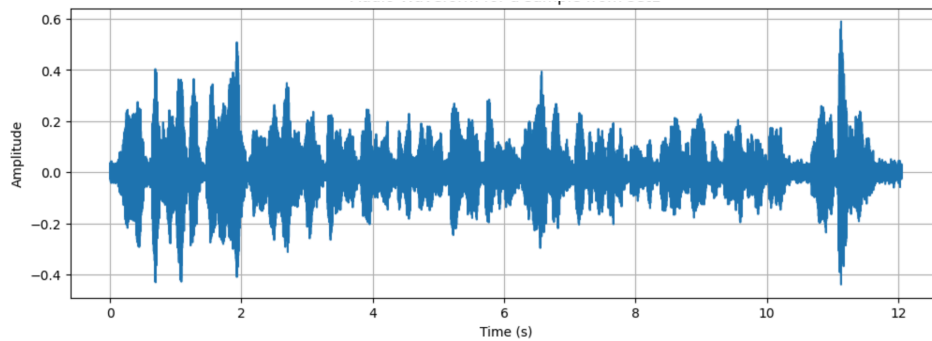


Figure 1. Raw speech waveform: amplitude plotted against time.

3.2 Short-Time Fourier Transform and Spectrograms

Because speech is a non-stationary signal, its frequency characteristics change continuously with time. To analyse speech effectively, the signal is divided into short, overlapping frames, typically 20-25 ms in duration with a stride of around 10 ms. Within each short frame, the speech signal can be assumed to be approximately stationary. The Short-Time Fourier Transform (STFT) is then applied to each frame to compute its frequency spectrum. Combining the spectra of all frames over time yields a two-dimensional time-frequency representation known as a spectrogram, in which the horizontal axis represents time, the vertical axis represents frequency, and intensity represents signal energy.

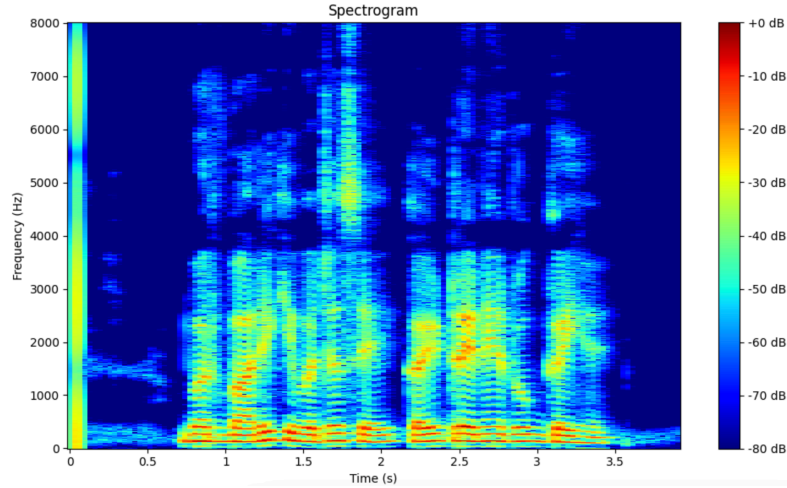


Figure 2. Standard (linear-frequency) spectrogram.

A standard spectrogram is usually represented using the magnitude or power spectrum. However, because the human auditory system perceives sound frequencies non-linearly, Mel spectrograms are widely used in speech recognition tasks. A Mel spectrogram is generated by passing the STFT output through a bank of Mel-scaled filters that allocate more resolution to lower frequencies and less resolution to higher frequencies, closely mirroring human hearing perception. This transformation reduces dimensionality while preserving perceptually important speech information.

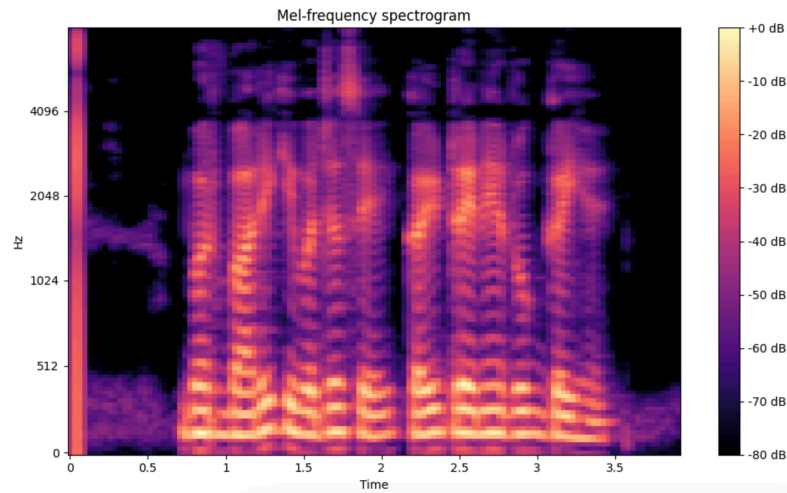


Figure 3. Mel-frequency spectrogram.

To further improve the representation, the logarithm of the Mel spectrogram values is computed, producing a Log-Mel spectrogram. Logarithmic scaling compresses the dynamic range of the signal and better reflects how humans perceive loudness. Log-Mel spectrograms are highly effective as input features for modern deep-learning-based ASR systems because they capture both temporal and perceptual frequency information in a compact and robust form.

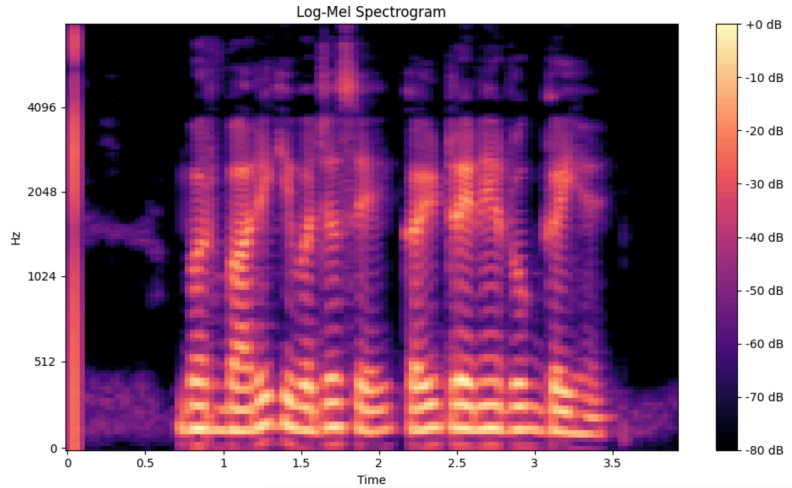


Figure 4. Log-Mel spectrogram.

3.3 Mel-Frequency Cepstral Coefficients (MFCCs)

MFCCs are compact, information-rich features derived from the Mel spectrum. After computing a Mel filterbank, the log-energy in each filter band is obtained, and the Discrete Cosine Transform (DCT) is applied to decorrelate the coefficients. Typically, 13 cepstral coefficients plus their first and second derivatives (delta and delta-delta) are used, yielding a 39-dimensional feature vector per frame. MFCCs dominated classical ASR pipelines for decades; modern deep learning systems often bypass hand-crafted features and learn representations directly from raw waveforms or Log-Mel spectrograms.

4. Evolution of ASR Systems

4.1 Classical Era: GMM-HMM (1980s-2012)

Early ASR systems relied on Gaussian Mixture Models (GMMs) and Hidden Markov Models (HMMs) to model speech sounds and their temporal structure. These systems required extensive feature engineering and separate acoustic, pronunciation, and language models. Although effective for their time, they struggled with variation in speakers, accents, and noisy environments.

4.2 Deep Learning Transition (2012-2016)

The introduction of Deep Neural Networks (DNNs) significantly improved speech recognition accuracy by learning better acoustic representations from data. DNNs gradually replaced GMMs in hybrid DNN-HMM systems, reducing the need for manual feature engineering and achieving lower error rates across many speech tasks.

4.3 End-to-End Learning (2016-2020)

End-to-end ASR models simplified the speech recognition pipeline by directly mapping audio signals to text using a single neural network. Approaches such as Connectionist Temporal Classification (CTC), attention-based models, and Transformer architectures reduced system complexity and improved performance given large-scale training data.

4.4 Self-Supervised and Foundation Models (2020-Present)

Modern ASR systems use self-supervised learning to learn speech representations from vast amounts of unlabelled audio. Foundation models such as wav2vec 2.0 and Whisper achieve strong multilingual and robust performance, requiring less labelled data and adapting well to diverse speech conditions.

4.5 Paradigm Comparison Summary

ASR technology has evolved from rule-based statistical systems (GMM-HMM) to deep learning models, then to end-to-end architectures, and finally to large self-supervised foundation models. Each stage has improved accuracy, scalability, and robustness while reducing dependence on hand-crafted components and labelled training data.

5. Indian Speech Recognition Models

Several dedicated efforts have produced ASR models for Indian languages. Table 2 provides an overview of the most prominent systems.

Table 2. Overview of prominent Indic ASR models

Model	Architecture	Language Coverage	Access
IndicConformer	Multilingual Transformer (Conformer)	All 5 languages (multilingual)	Open (MIT, Hugging Face)
IndicWhisper	Fine-tuned Whisper	All 5 languages (Indic-focused)	Open (MIT, Hugging Face)
IndicWav2Vec	Wav2Vec2-based	All 5 languages (22 languages total)	Open (MIT, Hugging Face)
Vaani-FastConformer	FastConformer (NVIDIA NeMo), streaming-capable	Hindi, Telugu	Open (Hugging Face)
BharatGen Shrutam	e-Branchformer (ESPnet)	Hindi (monolingual)	Open (Hugging Face: bharatgenai/Shrutam-HindiASR-1.0)
BharatGen Shrutam-2	Conformer encoder + MoE Projector (SMEAR-based soft routing, 8 experts) bridging to a frozen pretrained LLM decoder	12 languages: Hindi, Marathi, Tamil, Telugu, Malayalam, Kannada, Odia, Bengali, Urdu, Assamese, Gujarati, Punjabi	Open (Hugging Face: bharatgenai/Shrutam-2; BharatGen non-commercial license)
SarvamAI (Saaras v3)	Proprietary Transformer	22 scheduled Indian languages + English	Closed API (subscription key required)
Gnani Prisma v2.5	Not disclosed	23 languages (22 Indian + English)	Closed API (subscription key required)

AI4Bharat, a publicly funded research initiative at IIT Madras, has been the most prolific contributor to open-source Indic ASR; its models are fine-tuned versions of Wav2Vec2-XLS-R and Whisper trained on curated Indic corpora. Google's Universal Speech Model (USM) covers all five target languages but is accessible only via the Google Cloud Speech-to-Text API. Microsoft Azure Speech offers API access for Hindi, Telugu, and Tamil with competitive accuracy.

6. Indian Speech Audio Datasets

Curated, high-quality labelled speech data is the primary bottleneck for Indic ASR. Table 3 summarises the principal publicly available datasets used for training and evaluation.

Table 3. Principal publicly available Indic speech datasets

Dataset	Source	Languages	Description	Access
IndicSUPERB	AI4Bharat	22	Benchmark suite for speech processing tasks	Open
Shrutilipi	AI4Bharat	12	6,400+ hours of transcribed broadcast speech	Open
IndicTTS	IIT Madras	13	Combined TTS and ASR corpus	Open
MUCS 2021	IIT Madras	3 (Hindi, Marathi, Tamil)	Competition benchmark	Open
Common Voice (IN)	Mozilla	3 (Hindi, Telugu, Tamil)	Crowd-sourced speech	Open
Nirantar	AI4Bharat	22	3,240 hours of transcribed extempore/conversational speech across 208 Indian districts; continual-learning benchmark	Open
Sruti	AI4Bharat	1 (Bhojpuri)	Benchmark for rural Bhojpuri women speakers (72 minutes, 444 utterances) focused on healthcare, finance, and government domains	Open
Mahadhwani	AI4Bharat	22	Large-scale pre-training corpus for multilingual ASR covering all 22 scheduled Indian languages	Open
IndicVoices	AI4Bharat	22	23.7K hours of spontaneous, read, and conversational speech from 51K speakers across 400+ districts	Open
Lahaja	AI4Bharat	1 (Hindi)	Multi-accent Hindi benchmark from 132 speakers spanning 83 districts and 19 native language backgrounds	Open
Svarah	AI4Bharat	1 (English)	9.6 hours of Indian-accented English from 117 speakers across 65 districts and 19 language backgrounds	Open
Vistaar	AI4Bharat	12	59-benchmark evaluation suite spanning Kathbath, FLEURS, Common Voice, IndicTTS, MUCS and GramVaani	Open
Kathbath	AI4Bharat	12	1,684 hours of labelled read speech from 1,218 speakers across 203 districts	Open
BhasaAnuvaad	AI4Bharat	14	Largest Indic speech-translation dataset: 44,400+ hours of speech and 17M text segments (Indic-English)	Open (CC BY 4.0)

6.1 The Vistaar Benchmark in Detail

Because the Vistaar benchmark suite underpins much of the SOTA comparison and results reported in this study, its composition is detailed further here: the language coverage of its constituent datasets, the scale of its associated training corpus, and reference performance figures reported by Bhogale et al. (2023) for publicly available models.

6.1.1 Benchmark Coverage by Dataset and Language

Table 3.1 shows which of the seven constituent datasets provide evaluation benchmarks for each of the twelve languages covered by Vistaar, comprising 59 benchmarks in total.

Table 3.1. The Vistaar benchmark set — coverage across 59 benchmarks over 12 Indian languages

Dataset	bn	gu	hi	kn	ml	mr	or	pa	sa	ta	te	ur
Kathbath	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Kathbath-Hard	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
FLEURS	✓	✓	✓	✓	✓	✓	✓	✓		✓	✓	✓
CommonVoice	✓		✓		✓	✓	✓	✓		✓		✓
IndicTTS	✓	✓	✓	✓	✓	✓	✓			✓	✓	
MUCS		✓	✓			✓	✓			✓	✓	
Gramvaani			✓									

6.1.2 Vistaar-Train Corpus

Table 3.2 shows the number of hours of training data contributed by each source dataset, by language, in the Vistaar-Train corpus used to fine-tune models such as IndicWhisper.

Table 3.2. The Vistaar-Train dataset — hours of data across constituent datasets and languages

Dataset	bn	gu	hi	kn	ml	mr	or	pa	sa	ta	te	ur	Total
Shrutilipi	443	460	1620	459	359	1015	601	94	27	794	390	193	6,457
Kathbath	103	116	137	153	134	172	99	124	102	172	142	74	1,527
CommonVoice	64	-	13	-	1	19	2	2	-	226	-	46	373
NPTEL	44	59	157	80	60	68	-	-	-	206	183	-	857
IISc-MILE	-	-	-	347	-	-	-	-	-	150	-	-	497
MUCS	-	40	95	-	-	94	95	-	-	40	40	-	403
IndicTTS	20	21	20	19	18	19	19	-	-	20	36	-	192
IITB-MSR	-	-	-	-	-	109	-	-	-	-	-	-	109
Gramvaani	-	-	100	-	-	-	-	-	-	-	-	-	100
FLEURS	11	9	7	8	10	12	3	6	-	9	8	7	90
Vakyansancayah	-	-	-	-	-	-	-	-	78	-	-	-	78
Google TTS	5	8	-	8	6	3	-	-	-	7	6	-	43
IIITH-IndicSpeech	2	-	1	2	2	2	-	-	-	1	2	-	12
Total	691	712	2,150	1,077	590	1,513	819	226	207	1,625	806	320	10,736

6.1.3 Reference Comparison: Publicly Available Models on the Hindi Subset

Table 3.3 reproduces the reported WER comparison of publicly available ASR models on the Hindi subset of the Vistaar benchmark. IndicWhisper reports the lowest WER across every constituent dataset.

Table 3.3. Comparison of publicly available models on the Hindi subset of the Vistaar benchmark (WER %)

Model	Kathbath	Kathbath-Hard	FLEURS	CommonVoice	IndicTTS	MUCS	Gramvaani	Average
Google STT	14.3	16.7	19.4	20.8	18.3	17.8	59.9	23.9
IndicWav2Vec	12.2	16.2	18.3	20.2	15.0	22.9	42.1	21.0
Azure STT	13.6	15.1	24.3	14.6	15.2	15.1	42.3	20.0
NVIDIA-medium	14.0	15.6	19.4	20.4	12.3	12.4	41.3	19.4
NVIDIA-large	12.7	14.2	15.7	21.2	12.2	11.8	42.6	18.6
IndicWhisper	10.3	12.0	11.4	15.0	7.6	12.0	26.8	13.6

6.1.4 Full Vistaar Benchmark Performance by Language

Table 3.4 reports the per-dataset, per-language WER of four publicly available models across the full Vistaar benchmark, as reported by Bhogale et al. (2023). “NA” indicates the API did not support the language at the time of evaluation; “-” indicates no benchmark exists for that dataset-language pair.

Table 3.4.1 Google STT — WER (%) across the Vistaar benchmark

Dataset	bn	gu	hi	kn	ml	mr	or	pa	sa	ta	te	ur	Avg
Kathbath	20.6	26.0	14.3	27.9	52.6	26.6	105.7	25.5	104.2	31.1	34.3	17.6	40.5
Kathbath-Hard	22.4	28.4	16.6	31.4	53.6	28.4	105.4	29.2	104.4	32.8	35.9	20.2	42.4
CommonVoice	26.1	-	20.8	-	54.0	27.5	101.7	29.2	-	42.1	-	26.9	41.0
FLEURS	27.3	34.7	19.4	33.8	41.0	33.2	100.8	44.0	-	36.6	40.1	28.4	39.9
IndicTTS	37.0	33.6	18.3	32.8	38.3	21.8	104.7	-	-	31.3	55.4	-	41.5
MUCS	-	59.7	17.8	-	-	21.4	103.5	-	-	27.8	46.4	-	46.1
Gramvaani	-	-	59.9	-	-	-	-	-	-	-	-	-	59.9
Average	26.7	36.5	23.9	31.5	47.9	26.5	103.6	32.0	104.3	33.6	42.4	23.3	44.3

Table 3.4.2 Azure STT — WER (%) across the Vistaar benchmark

Dataset	bn	gu	hi	kn	ml	mr	or	pa	sa	ta	te	ur	Avg
Kathbath	16.7	22.4	13.6	23.6	43.3	20.9	NA	NA	NA	28.1	22.8	NA	-
Kathbath-Hard	18.4	26.7	14.9	25.9	45.7	NA	NA	NA	NA	30.0	24.6	NA	-
CommonVoice	15.0	-	14.6	-	47.5	19.6	NA	NA	NA	39.9	-	NA	-
FLEURS	29.9	34.8	24.3	31.0	39.2	37.2	NA	NA	NA	34.9	32.3	NA	-
IndicTTS	21.2	50.5	15.2	25.3	33.5	31.6	NA	NA	NA	31.2	61.1	NA	-
MUCS	-	23.7	15.1	-	-	27.5	NA	NA	NA	25.1	16.1	NA	-
Gramvaani	-	-	42.3	-	-	-	NA	NA	NA	-	-	NA	-
Average	20.2	31.6	20.0	26.4	41.8	27.4	NA	NA	NA	31.5	31.4	NA	-

Table 3.4.3 IndicWav2Vec — WER (%) across the Vistaar benchmark

Dataset	bn	gu	hi	kn	ml	mr	or	pa	sa	ta	te	ur	Avg
Kathbath	14.3	17.4	12.2	20.8	37.8	20.4	22.2	15.1	37.0	27.4	26.9	16.0	22.3
Kathbath-Hard	18.1	21.5	16.2	26.4	42.8	24.0	28.3	18.2	43.3	33.0	31.2	19.4	26.9
FLEURS	28.4	22.7	18.3	22.0	40.4	25.1	40.5	34.8	-	37.7	30.4	32.2	30.2
CommonVoice	31.3	-	20.2	-	67.0	27.0	37.9	19.7	-	42.7	-	32.7	34.8
IndicTTS	28.1	22.7	15.0	20.1	40.5	18.6	24.4	-	-	28.0	32.3	-	25.5
MUCS	-	32.6	22.9	-	-	34.7	30.1	-	-	37.7	30.4	-	31.4
Gramvaani	-	-	42.1	-	-	-	-	-	-	-	-	-	42.1
Average	24.0	23.4	21.0	22.3	45.7	25.0	30.6	21.9	40.2	34.4	30.2	25.1	28.7

Table 3.4.4 IndicWhisper — WER (%) across the Vistaar benchmark

Dataset	bn	gu	hi	kn	ml	mr	or	pa	sa	ta	te	ur	Avg
Kathbath	16.6	17.8	10.1	19.3	34.8	19.9	24.7	16.9	45.6	24.2	25.0	11.9	22.2
Kathbath-Hard	19.4	20.6	11.9	22.2	38.4	22.1	29.1	19.7	50.5	27.5	27.8	14.7	25.3
CommonVoice	24.7	-	14.2	-	44.5	22.8	35.2	22.4	-	29.2	-	31.7	28.1
FLEURS	20.9	23.5	10.5	18.6	22.6	20.5	32.9	23.1	-	25.2	25.4	19.2	22.0
IndicTTS	18.8	19.1	8.1	13.2	21.4	11.4	15.0	-	-	17.2	33.8	-	17.6
MUCS	-	33.2	17.1	-	-	12.8	27.5	-	-	28.3	32.1	-	25.2
Gramvaani	-	-	24.9	-	-	-	-	-	-	-	-	-	24.9
Average	20.1	22.8	13.8	18.3	32.3	18.2	27.4	20.5	48.0	25.3	28.8	19.4	24.6

Consistent with Table 3.3, IndicWhisper achieves the lowest average WER among the four models on nearly every language, with the gap most pronounced on Sanskrit (sa) and Odia (or), where Google STT's WER exceeds 100% (indicating predictions far longer than the reference transcripts). Azure STT does not support Odia, Punjabi, or Sanskrit, and Gramvaani (a Hindi-only rural telephony benchmark) shows uniformly higher error rates across all models, reflecting the difficulty of noisy, low-resource telephony speech.

7. Top-3 State-of-the-Art Models per Language (as Reported in the Literature)

Table 4. Top-3 reported SOTA models by language

Language	#1	#2	#3	Notes
Hindi	IndicWhisper (AI4Bharat)	IndicConformer (RNNT)	Wav2Vec2-based models (MUCS/Shrutilipi fine-tuned)	The Vistaar benchmark reports IndicWhisper achieving the lowest WER in 39 of 59 benchmarks, an average reduction of 4.1 WER over compared systems; Hindi has the widest benchmark coverage of any Indic language (Kathbath, MUCS, LAHAJA, Common Voice, FLEURS).
Bengali	IndicWhisper	IndicConformer (RNNT)	Wav2Vec2 (OpenSLR-trained)	Bengali benefits from relatively larger labelled corpora (OpenSLR, Shrutilipi) compared with the Dravidian languages, which the literature associates with its comparatively lower error rates.
Marathi	IndicWhisper	IndicConformer (RNNT)	Wav2Vec2 (MUCS-trained)	Marathi's moderate agglutination places it between Hindi/Bengali and the Dravidian languages in reported difficulty.
Telugu	IndicConformer (RNNT)	IndicWhisper	Vaani-FastConformer	Telugu and Tamil are consistently flagged in the literature as harder due to agglutinative morphology and comparatively smaller labelled training sets.
Tamil	IndicConformer (RNNT)	IndicWhisper	Vaani-FastConformer	Cross-lingual ASR studies note that Tamil ASR systems trained on hundreds of hours of data still show the largest generalisation gaps among Indic languages when applied cross-lingually, reflecting how challenging the language is even in-domain.

8. Evaluation Methodology

8.1 Metrics: Word Error Rate and Character Error Rate

Word Error Rate (WER) is the primary metric for ASR evaluation. It is defined as the minimum edit distance (insertions, deletions, substitutions) between the hypothesis and reference word sequences, normalised by the number of reference words:

$$WER = (S + D + I) / N \times 100\%$$

where S = substitutions, D = deletions, I = insertions, and N = the reference word count. For morphologically complex languages such as Tamil and Telugu, Character Error Rate (CER) is also reported, since the character provides a finer-grained signal than the word unit in these languages.

8.2 Hardware and Inference Configuration

All ASR models were evaluated in 32-bit floating-point mode using cloud-based notebook environments specifically Google Colab and Kaggle each equipped with a single NVIDIA T4 GPU (16 GB VRAM). Experiments were conducted in these cloud-based environments to assess the practical feasibility of deploying speech recognition systems on widely accessible, cost-effective hardware. Batch size was varied across experiments depending on model and memory constraints, rather than fixed, to accommodate differences in model architecture while remaining within the available GPU memory. No external language-model re-scoring was applied, in order to evaluate the intrinsic acoustic-modelling capability of each system under consistent inference settings.

9. Results and Discussion

9.1 Per-Language WER and CER Results

Table 5 presents the WER and CER of the evaluated models for each of the five languages, grouped by test set.

Table 5.1 Hindi — WER and CER results

Model	Source	WER (%)	CER (%)	Test Set	Test Set Size	Rank
Vaani-FastConformer-Hindi	Open	15.11	7.09	IndicVoices-Hindi	5,530	1
IndicConformer-600M-multilingual	Open	15.3 (RNNT) / 16.6 (CTC)	7.2 (RNNT) / 7.3 (CTC)	IndicVoices-Hindi	5,530	2
IndicWav2Vec-Hindi	Open	38.6	22.5	IndicVoices-Hindi	5,530	3
Gnani Prisma v2.5	Closed	8.78	3.26	Kathbath-Hindi	1,929	1
IndicConformer-600M-multilingual	Open	9.44 (RNNT) / 9.74 (CTC)	3.07 (RNNT) / 3.17 (CTC)	Kathbath-Hindi	1,929	2
IndicWhisper-Hindi	Open	10.23	3.65	Kathbath-Hindi	1,929	3
IndicWav2Vec-Hindi	Open	12.49	4.3	Kathbath-Hindi	1,929	4
Vaani-FastConformer-Hindi	Open	12.49	4.3	Kathbath-Hindi	1,929	5
Sarvam Saaras v3	Closed	14.78	4.59	Kathbath-Hindi	1,929	6

Table 5.2 Bengali — WER and CER results

Model	Source	WER (%)	CER (%)	Test Set	Test Set Size	Rank
IndicConformer-600M-multilingual	Open	13.49 (RNNT) / 14.23 (CTC)	5.36 (RNNT) / 5.41 (CTC)	IndicVoices-Bengali	3,906	1
IndicWav2Vec-Bengali	Open	46.7	21.2	IndicVoices-Bengali	3,906	2
IndicConformer-600M-multilingual	Open	13.2 (RNNT) / 13.58 (CTC)	3.82 (RNNT) / 3.94 (CTC)	Kathbath-Bengali	1,783	1
Gnani Prisma v2.5	Closed	13.94	4.79	Kathbath-Bengali	1,783	2
IndicWhisper-Bengali	Open	19.22	5.51	Kathbath-Bengali	1,783	3
Sarvam Saaras v3	Closed	21.21	5.07	Kathbath-Bengali	1,783	4
IndicWav2Vec-Bengali	Open	22.69	4.68	Kathbath-Bengali	1,783	5

Table 5.3 Marathi — WER and CER results

Model	Source	WER (%)	CER (%)	Test Set	Test Set Size	Rank
IndicConformer-600M-multilingual	Open	14.9 (RNNT) / 16 (CTC)	5.61 (RNNT) / 5.69 (CTC)	IndicVoices-Marathi	3,552	1
Gnani Prisma v2.5	Closed	16.89	5.72	Kathbath-Marathi	1,631	1
IndicConformer-600M-multilingual	Open	17.14 (RNNT) / 17.61 (CTC)	5.5 (RNNT) / 5.62 (CTC)	Kathbath-Marathi	1,631	2
IndicWhisper-Marathi	Open	20.25	6.52	Kathbath-Marathi	1,631	3
Sarvam Saaras v3	Closed	20.46	5.96	Kathbath-Marathi	1,631	4

Table 5.4 Telugu — WER and CER results

Model	Source	WER (%)	CER (%)	Test Set	Test Set Size	Rank
IndicConformer-600M-multilingual	Open	26.28 (RNNT) / 27.86 (CTC)	8.42 (RNNT) / 8.53 (CTC)	IndicVoices-Telugu	3,295	1
Vaani-FastConformer-Telugu	Open	27.31	8.77	IndicVoices-Telugu	3,295	2
ARTPARK-IISc/whisper-medium-vaani-telugu	Open	61.69	25.69	IndicVoices-Telugu	3,295	3
IndicConformer-600M-multilingual	Open	21.37 (RNNT) / 21.92 (CTC)	3.55 (RNNT) / 3.63 (CTC)	Kathbath-Telugu	1,492	1
Sarvam Saaras v3	Closed	24.8	3.7	Kathbath-Telugu	1,492	2
Vaani-FastConformer-Telugu	Open	23.29	4.06	Kathbath-Telugu	1,492	3
Gnani Prisma v2.5	Closed	25.53	7.55	Kathbath-Telugu	1,492	4
IndicWhisper-Telugu	Open	36.9	12.9	Kathbath-Telugu	1,492	5

Table 5.5 Tamil — WER and CER results

Model	Source	WER (%)	CER (%)	Test Set	Test Set Size	Rank
IndicConformer-600M-multilingual	Open	30.7 (RNNT) / 34.7 (CTC)	9.1 (RNNT) / 10.1 (CTC)	IndicVoices-Tamil	5,276	1
IndicConformer-600M-multilingual	Open	23.37 (RNNT) / 24.59 (CTC)	3.95 (RNNT) / 4.29 (CTC)	Kathbath-Tamil	1,642	1
IndicWhisper-Tamil	Open	24.9	4.47	Kathbath-Tamil	1,642	2

Model	Source	WER (%)	CER (%)	Test Set	Test Set Size	Rank
Sarvam Saaras v3	Closed	28.65	4.74	Kathbath-Tamil	1,642	3
Gnani Prisma v2.5	Closed	24.96	5.51	Kathbath-Tamil	1,642	4

For Hindi and Bengali, WER works reasonably well since these languages have relatively simple morphology and stable word boundaries. For Marathi, Telugu, and especially Tamil, which are more agglutinative, stacking suffixes and case markers onto root words CER becomes more important, since a single suffix error can mark an entire word as incorrect under WER even when the model recognised most of it correctly.

11. Conclusion and Future Work

This project establishes a reproducible benchmark for ASR across India's five most spoken languages, evaluated on both natural spontaneous speech (IndicVoices) and clean read speech (Kathbath). Key observations:

- IndicConformer-600M-Multilingual (RNNT) is the strongest and most consistent open-source model. RNNT decoding outperforms CTC in every single language/dataset pairing measured, with the largest gaps on Tamil (30.7% vs 34.7% WER, IndicVoices) and Telugu (26.28% vs 27.86% WER, IndicVoices).
- On Kathbath, IndicConformer RNNT beats both commercial APIs in 3 of 5 languages — Bengali (13.2% WER / 3.82% CER vs. Gnani's 13.94%/4.79%), Telugu (21.37%/3.55% vs. Sarvam's 24.8%/3.7% and Gnani's 25.53%/7.55%), and Tamil (23.37%/3.95%, best of all five models tested, ahead of IndicWhisper, Sarvam, and Gnani). This is a meaningfully different picture than "commercial beats open-source" — it mostly doesn't, once RNNT decoding is used.
- Hindi and Marathi are the two languages where a commercial API (Gnani) edges out IndicConformer on WER — Hindi (8.78% vs. 9.44%) and Marathi (16.89% vs. 17.14%) — though IndicConformer still wins or ties on CER in both cases (3.07% vs. 3.26% Hindi; 5.5% vs. 5.72% Marathi). Gnani's Hindi result (8.78% WER) is the single best number in the whole benchmark.
- On IndicVoices (harder, spontaneous speech), Vaani-FastConformer-Hindi is the only model that edges out IndicConformer, on Hindi specifically (15.11% vs. 15.3% WER, RNNT) — making it a reasonable pick for Hindi-only, low-latency deployments. On Telugu, Vaani-FastConformer-Telugu (27.31%/8.77%) trails IndicConformer RNNT (26.28%/8.42%) by less than a point, making both viable depending on latency/streaming requirements.
- IndicWav2Vec is not production-viable. It trails IndicConformer badly on every language tested — 38.6% vs. 15.3% WER on Hindi (IndicVoices) and 46.7% vs. 13.49% on Bengali (IndicVoices) — and remains behind even on the easier Kathbath set.
- Tamil is the hardest language in the benchmark (best result: 23.37% WER on Kathbath, 30.7% on IndicVoices), roughly double Bengali's error rate (13.2%/13.49%), pointing to lower training-data coverage and/or higher acoustic/script complexity. Bengali is the easiest, with the strongest results across nearly every model tested.

Overall, for multilingual Indic ASR, IndicConformer-600M with RNNT decoding is the recommended open-source baseline, while Gnani Prisma v2.5 is the best-performing commercial option tested in this benchmark.

12. References

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